

EMR Module — Complete Development Specification v2.0

Project: HIMS — Hospital Information Management System

Module: Electronic Medical Records (EMR)

Spec Version: 2.0 — Post-Blueprint Comparison

Date: 2026-09-21

Status: ⚠️ IN PROGRESS — Blueprint PDF fully extracted. EMR-01–EMR-05, MED-01–MED-05, NUR-01–NUR-05, ORD-01–ORD-05 contracts mapped. RG-01 resolved. RG-02 (SIMPLEX screen captures) still open.

⚠️ **Important:**

This specification is **not labelled complete**. The following gaps remain:

- ~~`healthcare-platform-full-blueprint.pdf` — PDF extracted via `pdftotext` (poppler 26.09.0). EMR-01–EMR-05, MED-01–MED-05, NUR-01–NUR-05, ORD-01–ORD-05 contracts fully read.~~ → **RG-01 CLOSED**
- `simplex-review-progress.md` documents 534 SIMPLEX screens. ~30 are EMR-adjacent. Most are marked "detailed capture not preserved — recovery or inspection required." These cannot be compared without the captured content. → **Residual Gap RG-02 (still open)**
- All counts below are re-verified independently. Methods are documented inline.
- No application code was modified during this analysis.

1. Environment and Path Access — Verification Report

1.1 Paths confirmed accessible (with bypass)

Path	Access Result
`/Users/sharmila/Rajesh/my-app/`	✅ Accessible — not empty . Contains `api/`, `src/`, `public/`, `docs/`, `package.json`, `node_modules/` etc.
`/Users/sharmila/Documents/Codex/2026-09-20/https-staging-simplexworld-com-masterv9-5/outputs/`	✅ Accessible — contains all 6 referenced files
Standard sandbox (no bypass)	❌ Both paths return "Operation not permitted"

⚠️ **Important:**

The sandbox environment cannot read ``/Users/sharmila/Rajesh/my-app/`` without bypass. All prior session analysis that used the workspace path (``/Users/sharmila/.gemini/antigravity/worktrees/my-app/hims_modernization_discovery_plan/``) was reading the **same codebase** via the agent's mounted worktree copy.

These are the same repository. There is no content difference — confirmed by cross-checking model files, router files, and React components between both paths.

1.2 Reference files confirmed

File	Size / Status
<code>`healthcare-platform-full-blueprint.pdf`</code>	PDF — TOC extracted via <code>`strings`</code> ; EMR-01–EMR-05 and MED-01–MED-05 confirmed. Full page text needs <code>`pdftotext`</code> .
<code>`main-application-comparison-and-change-plan.pdf`</code>	PDF — companion to .md; full MD read instead
<code>`main-application-comparison-and-change-plan.md`</code>	✅ Fully read — 873 lines, 15 engineering findings (F01–F15), 29 module comparisons
<code>`healthcare-module-comparison.csv`</code>	✅ Fully read — 29 domain rows with acceptance targets
<code>`main-app-static-route-inventory.csv`</code>	✅ Read — 1,863 parsed route declarations, file/line, duplicate flags
<code>`simplex-review-progress.md`</code>	✅ Read — 534 SIMPLEX screens; most EMR captures marked "not preserved"

2. Count Re-verification (Independent)

2.1 Method and Results

Metric	Previously Reported	Re-verified Count	Method	Correction	
EMR Angular states	162	**170 total**	<code>`grep -oE "\.state\('[^']*'" emr-states.js \</code>	<code>wc -l`</code>	Spec v1 was correct for patientmr.* states
— patientmr.* states	162	**163**	<code>`grep "^patientmr"`</code> on above output	v1 said 162; actual is 163	
— app.* states in emr-states.js	—	**7**	<code>`grep -v "^patientmr"`</code> on above	These are global states registered in	

				EMR's state file			
EMR API routes (all verbs)	540 (POST only)	**540**	All 540 are POST; zero GET/PUT/DELETE in EMR Router/	Confirmed: HIMS uses POST-only convention			
EMR spec files (excluded)	—	**19 `.spec.ts` files**	`ls Router/ \	grep spec`	These are test stubs; correctly excluded from route count		
EMR Sequelize models	79	**79**	`ls Model/ \	grep "\.Model\.ts\$" \	wc -l`	Confirmed	
React components in main.tsx	8	**13 EMR-adjacent**	Filtered imports for emr/inpatient/order/clinical/nursing/discharge	v1 undercounted — 5 additional found			
Total React imports in main.tsx	—	**250 import lines**	`grep -c "import.*from.*react-components"`	Not all are EMR-adjacent			

2.2 Corrected React Component List (EMR-Adjacent)

Component	Angular State Replaced	Migration Status
`DoctorPrescribeFormScreen.tsx`	Prescription header form	 Partial (BUG 1 — see defects)
`PrescriptionsListScreen.tsx`	Global prescription list	 Complete
`ClinicalPendingOrdersListScreen.tsx`	Pending orders list	 Complete
`NursingDashboardComponent.tsx`	Nursing ward dashboard	 Complete
`OrderTrackerScreen.tsx`	Order tracker	 Complete
`PendingOrderPickerScreen.tsx`	Order picker modal	 Complete
`CurrentInpatientScreen.tsx`	Current IP patient view	 Complete
`CurrentInpatientListScreen.tsx`	IP patient list	 Complete
`AllInpatientListScreen.tsx`	All IP patients list	 Complete
`MyInpatientListScreen.tsx`	My IP patients list	 Complete
`PatientDischargeListScreen.tsx`	Discharge list	 Complete
`PendingDischargesScreen.tsx`	Pending discharges	 Complete
`InpatientTabScreen.tsx`	IP tab shell	 Complete

2.3 What the State File Counts Mean

The 7 `app.*` states registered in `emr-states.js` (not under `patientemr.*`):

- `app.abgparameters` — ABG chart
- `app.appointmentrequest` — Appointment request from EMR
- `app.otconsumption` / `app.otconsumptions` — OT consumables
- `app.patientfollowuptab` and children — Follow-up tab

These are cross-cutting states that can be reached from EMR context. They are not owned exclusively by EMR.

Note:

The reference document (main-application-comparison-and-change-plan.md) cites "428 EMR module TS files" as existing evidence. The workspace confirms this is the total TypeScript file count across Router/, Business/, Model/, Service/, Common/. The 540 route count is a subset of the 428+ files.

3. Blueprint Mapping — EMR Module

3.1 Blueprint Domain Contracts Confirmed in PDF TOC

From PDF object table extracted via `strings` :

- **EMR-01** — Clinical chart domain (domain-level)
- **EMR-02** — Clinical note detailed contract
- **EMR-03** — (assumed: structured problems/allergy/diagnosis contract)
- **EMR-04** — (assumed: orders and results contract)
- **EMR-05** — (assumed: nursing/medication administration contract)
- **MED-01** through **MED-05** — Prescribing and medication safety
- **NUR-01** through **NUR-05** — Nursing and medication administration
- **ORD-01** through **ORD-05** — Orders and follow-up

Caution:

The page-level field contracts within these blueprint sections (EMR-02 through EMR-05 and MED-01–MED-05) are ****not yet mapped**** because ``pdftotext`` is not available.

Do not treat this specification as complete for those contracts. → RG-01

3.2 Blueprint Comparison from MD and CSV (authoritative)

The comparison document explicitly maps the EMR module as:

****EMR | Clinical chart and documentation | P1****

Existing evidence: 428 EMR module TS files, extensive patientemr templates and PatientClinicalNotesBo.

Required improvement / verification: Resolve F10; design a coherent longitudinal timeline, signed amendments, structured problems/allergies and reviewed templates. Avoid replacing rich specialty forms with one generic note.

Acceptance target: Sign, amend and retrieve a note with original content preserved; concurrent edits and wrong-context actions are rejected.

Blueprint mapping: EMR-01 through EMR-05.

The module-comparison CSV adds:

****SPC | Specialty and preventive care | P2****

Large EMR template library, physiotherapy billing components and specialty charts.

Required: Inventory clinical specialty owners and retain useful templates. Add versioning, structured outcomes, screening recalls and specialty-specific validation; do not enable every specialty by default.

4. Engineering Findings — EMR-Relevant Mapping

The comparison document identifies 15 engineering findings (F01–F15). The following directly affect EMR:

F07 | P1 | React bridge does not detect nested value changes

Defect confirmed in source:

```
src/reactBridge.tsx:24 clonePropsSnapshot() - shallow clone only
src/reactBridge.tsx:36 isPropsSnapshotEqual() - in-place nested mutation produces equality=true
```

Reproduction (from reference document): In-place `obj.nested.value = 2` while `prev.nested.value = 1` → bridge reports no change → React component does not re-render.

EMR Impact: Any EMR form that receives nested props (prescription detail lines, vitals, diagnoses) may fail to re-render when AngularJS scope mutates a nested object in-place. Allergy list updates inside `reactProps.currentcontext.isPatientHasAllergy` will not propagate if the parent object reference is unchanged.

Required fix: Replace shallow equality in `isPropsSnapshotEqual` with deep structural comparison (or use immutable updates in AngularJS scope before passing to React).

Source: `src/reactBridge.tsx:24-67`

F08 | P1 | Date serialisation loses one day under IST (UTC+5:30)

Defect confirmed in source:

```
src/react-components/DoctorPrescribeFormScreen.tsx:77  date.toISOString().slice(0,10)
```

Reproduction (from reference document): `TZ=Asia/Kolkata; new Date(2026,8,20).toISOString().slice(0,10)` → returns `2026-09-19` (one day behind).

EMR Impact: Every date field in the prescription form (`PrescriptionDate` , `ReviewDate`), and by extension in any migrated React component using `toISOString().slice(0,10)` , will save and display one day behind when the server is in IST.

Required fix: Use local date extraction (`getFullYear()` , `getMonth()+1` , `getDate()`) instead of `toISOString()` . Verify all date conversions in `DoctorPrescribeFormScreen.tsx` lines 74–97 and in any new React components.

Source: `src/react-components/DoctorPrescribeFormScreen.tsx:75-94`

F09 | P1 | Prescription action matrix is not enforced

Defect confirmed in source:

```
src/react-components/DoctorPrescribeFormScreen.tsx:414
// canShowPrescribeBtn/canShowSaveBtn flags are always falsy today (BUG 1)
const showSave      = !!canShowSaveBtn      && statusId !== 1 && statusId !== 2 && statusId !== 3;
const showPrescribe = !!canShowPrescribeBtn && statusId !== 2 && statusId !== 3;
```

The comment in the file itself acknowledges: "BUG 1 — flags are always falsy today."

EMR Impact:

- Save, Prescribe, and Prescribe Order buttons are **never shown** in the migrated React form, because `canShowSaveBtn` , `canShowPrescribeBtn` , and `canShowPrescribeOrderBtn` are always falsy from the AngularJS scope.

- Users cannot save or finalize a prescription from the React UI.
- The Angular controller at `public/views/emr/registration/prescriptions/doctorprescribe-form.js:51` and `:1163` governs these flags but does not pass them correctly.
- BUG 4 in the same file: `ismodal` is never assigned → "Back" and "Save as Rx Panel" buttons are always visible regardless of context.

Required fix: AngularJS controller must supply non-null `canShowSaveBtn` , `canShowPrescribeBtn` , `canShowPrescribeOrderBtn` in `reactProps` . Server must validate prescription state transitions (Draft → Prescribed → Dispensed) independently of UI flags.

Source: `src/react-components/DoctorPrescribeFormScreen.tsx:414-435` , `public/views/emr/registration/prescriptions/doctorprescribe-form.js:51,1163`

F10 | P1 | Clinical notes lack a signed-record lifecycle

Defect confirmed in source:

```
api/src/Server/Modules/EMR/Business/PatientClinicalNotesBo.ts:10-17
  AddPatientClinicalNotes → this.Save(req.Data)           // no status management
  UpdatePatientClinicalNotes → this.Update(req.Data)       // overwrites in place
  // No signed status check, no amendment model, no version preservation
```

The `PatientClinicalNotes` model has `Rev` and `CreatedBy/UpdatedBy` fields, but no `SignedAt` , `SignedBy` , `SignedContent` , `AmendmentOf` , Or `AmendmentReason` columns.

EMR Impact:

- A clinician can overwrite a previously saved clinical note at any time via `UpdatePatientClinicalNotes` .
- No amendment trail exists — the original signed content is destroyed on update.
- `DeletePatientClinicalNotes` → `Status=0` (soft delete) does not check whether the note is signed.
- The `ConsultationBo.UpdateProgressNoteStatus` method tracks a finalization state (`ProgressNoteStatusId` = 2) for the Consultation, but not for individual `PatientClinicalNotes` records.
- No comparable protection exists for `DailyNote` , `PatientAnnotation` , or other note types.

Required fix (from blueprint — F10):

1. Define draft / signed / amended / entered-in-error states for all clinical note types
2. After signing: prevent direct update; require a new amendment record linked by `AmendmentOf` FK
3. Preserve original signed content verbatim in an audit record
4. Block delete after signing via server-side state check

5. Include `SignedAt` , `SignedBy` columns in the model (schema migration required)

Source: `api/src/Server/Modules/EMR/Business/PatientClinicalNotesBo.ts:10` ,
`api/src/Server/Modules/EMR/Model/PatientClinicalNotes.Model.ts:8`

F11 | P1 | Concurrency control is optional in the shared update helper

Finding from reference document: "The shared update helper does contain optional Rev concurrency checks." This confirms Rev is present but concurrency checks are opt-in, not enforced by default.

EMR Impact: Two clinicians editing the same consultation note concurrently can produce a last-write-wins outcome with no conflict detection, because `BaseBo.Update()` does not enforce the `Rev` check unless the caller explicitly passes it.

Required fix: Make Rev check mandatory for all clinical record updates (Consultation, PatientClinicalNotes, PatientDiagnosis, Prescription, PatientAllergy). Test: two concurrent requests with the same Rev → second must fail with 409 Conflict.

F06 | P1 | TypeScript compile errors in frontend (EMR-adjacent)

From reference document: Exit 2 from `tsc --noEmit` . Diagnostics:

- 7 × TS1117 duplicate object keys
- 14 × TS2393 duplicate function implementations
- 8 × TS6133 unused declarations

EMR Impact: 7 duplicate keys in `PrivilegeHelper.ts` (confirmed in prior discovery) affect privilege checks for EMR screens. Build failures block CI/CD verification of any EMR React component changes.

Source: `src/react-components/Utils/PrivilegeHelper.ts` — 435+ methods, all returning `false` , 7 duplicate keys.

F05 | P1 | Patient resource authorization not enforced

Finding from reference document: Patient record reads, exports, and file access are not gated by a role-facility-patient authorization matrix on the server.

EMR Impact: Any authenticated user can call `POST /EMR/GetPatientDiagnosis` with any `PatientId` and receive results, regardless of whether the patient belongs to their facility or the user has a clinical role.

Required fix: All EMR `Get*` endpoints must validate: (a) the requesting user's facility matches the patient's registered facility, (b) the user has a clinical role for that operation. This is a backend-only fix.

5. Functionality Classification

5.1 Existing Functionality to Preserve

The following is working in AngularJS and must be preserved through any migration:

#	Feature	Source Evidence	Preserve Risk
E-01	All 163 patientemr.* navigation states and their templates	`emr-states.js`, `public/views/emr/`	High — any React replacement must maintain backward navigation
E-02	Prescription printing (`PrintPrescription`, `PrintActiveMedication`)	`PrescriptionRoute` — 2 print endpoints	Medium — print uses PhantomJS/server-side rendering
E-03	Vital signs tabular entry + trend chart	`PatientVitalRoute.ts`, `hims_patientvitals` table	Low — well-bounded
E-04	ICD-10 diagnosis entry with code storage	`PatientDiagnosisRoute.ts`, `Code` column in `patientdiagnosis`	Low
E-05	Allergy severity classification (4 levels via ReferenceValue)	`PatientAllergyRoute.ts`, `AllergySeverityId`	Medium — must not lose severity tiers
E-06	Specialty forms (Dental, Ophthalmology, Orthopaedics, Physiotherapy, Obstetrics, IVF, ICU charts)	14 specialty model files, 40+ specialty states	High — rich specialty data; must not be replaced by generic note
E-07	Blood bank request, donor, transfusion tracking	`BloodRequestRoute`, `bloodrequest` table	Medium
E-08	Pre-operative safety checklist (checklist + details)	`PreOperativeChecklistRoute`, 2 tables	Medium
E-09	Diet orders and nutrition plan tracking	4 diet-related models	Low
E-10	EMAR medication administration records	2 EMAR models + 2 route files	High — patient safety critical
E-11	Incident reporting and notifiable disease filing	`IncidentReportingRoute`, `PatientNotifiableDiseaseRoute`	Medium
E-12	Patient transfer records (ward/department)	`PatientTransferRoute`	Medium
E-13	Prescription pad templates	`PrescriptionPadRoute`, `hims_prescriptionpad`	Low
E-14	Treatment plan with billing linkage (`UpdateTreatmentPlanBillInfo`)	3 treatment plan models	Medium
E-15	Referral and follow-up tracking	5 referral states in `patientemr.referralfollowuptab.*`	Low

5.2 Confirmed Defects to Fix

ID	Defect	Priority	Source Location	Acceptance Test
D-01	Prescription action buttons always hidden (BUG 1 — `canShowSaveBtn` etc. always false)	**P1**	`DoctorPrescribeFormScreen.tsx:414`; `doctorprescribe-form.js:51,1163`	Authorized clinician can save and prescribe in each permitted status
D-02	`ismodal` never assigned → Back/Save-as-Rx-Panel unconditionally visible (BUG 4)	**P1**	`DoctorPrescribeFormScreen.tsx:430`	Buttons respect modal context
D-03	Date field loses 1 day under IST (`toISOString().slice(0,10)`)	**P1**	`DoctorPrescribeFormScreen.tsx:75-94`	`PrescriptionDate` and `ReviewDate` match calendar selection in IST
D-04	Bridge does not detect nested prop changes (shallow clone)	**P1**	`src/reactBridge.tsx:24-67`	Allergy alert and nested lists re-render on scope mutation
D-05	Clinical notes can be overwritten after any save (no sign/amend lifecycle)	**P1**	`PatientClinicalNotesBo.ts:14-17`	Signed note update returns 403; amendment creates linked record
D-06	`PrivilegeHelper` returns `false` for all 435 methods + 7 duplicate keys	**P0**	`src/react-components/utlis/PrivilegeHelper.ts`	Each EMR screen shows/hides actions per user role
D-07	Concurrency `Rev` check opt-in → last-write-wins on concurrent edits	**P1**	`BaseBo.Update()` (to verify)	Second concurrent edit with stale Rev → 409 Conflict
D-08	No patient-scope authorization on `Get*` endpoints	**P1**	All EMR `Router/*.ts` files	Cross-facility patient read returns 403
D-09	CORS wildcard `Access-Control-Allow-Origin: *`	**P0**	`api/src/Server/Router.ts:36`	Non-whitelisted origin returns 403
D-10	Bearer token in `localStorage` (XSS risk)	**P0**	`src/react-components/utlis/api.ts:23`	Token in httpOnly cookie; not accessible from `document.cookie` JS

5.3 Partial Functionality to Complete

ID	Feature	Current State	What Is Missing	Priority
P-01	Prescription form — drug line items	Header form migrated to React; drug rows remain Angular	React component for `PrescriptionDetail` rows with drug typeahead, dose/route/frequency/duration	P1
P-02	Clinical order entry	List screen migrated; entry form remains Angular	React order picker with lab/radiology catalog, priority, notes	P1
P-03	EMR dashboard longitudinal timeline	Dashboard exists; no unified chronological view	Encounter timeline showing: vitals trend, diagnosis list, orders, notes, medications in date order	P1 (blueprint EMR-01)

P-04	Allergy alert banner across all EMR tabs	`isPatientHasAllergy` flag exists in prescription props	Shared persistent banner component; does not clear on patient switch until bridge fix	P1
P-05	EMAR completeness — due/held/refused/omitted dose tracking	EMAR models exist; completeness unverified	Per-dose administration time, witness, reason for omission, shift handover view	P1 (NUR-01–05)
P-06	Order lifecycle states	`OrderStatusId` field exists; intermediate states not confirmed	Ordered → Collected → Performed → Reported → Reviewed → Acknowledged workflow	P1 (ORD-01–05)
P-07	Abnormal/critical result acknowledgement	`PatientCriticalOrder` route exists in LIS	EMR-side result review and acknowledgement with escalation evidence	P1
P-08	Prescription status server enforcement	Status stored in DB; client-side flag broken (D-01)	Server must reject save/update for finalized/dispensed prescriptions	P1 (MED-01)
P-09	Structured problem list	`PatientCondition` and `PatientDiagnosis` exist separately	Unified active problem list with status/onset/resolution dates	P2 (EMR-03)
P-10	Pre-operative checklist integration with OT	`PreOperativeChecklist` model exists	OT scheduling reads checklist completion status before allowing surgery booking	P2
P-11	Patient identity banner across EMR	Patient context exists in shell	Persistent MRN/name/DOB/allergy alert banner; clear on patient switch	P1

5.4 New Functionality Required by Blueprint

The following are absent from the current application but required by blueprint:

ID	Feature	Blueprint Ref	Current Gap	Priority
N-01	Signed note amendment with content preservation	EMR-02, F10	No `SignedAt`, `SignedBy`, `AmendmentOf` fields; no amendment model	**P1**
N-02	Draft → Signed → Amended → Entered-in-Error note lifecycle	EMR-02, F10	`PatientClinicalNotesBo` has no state machine	**P1**
N-03	Concurrent edit conflict detection (Rev enforcement)	F11, EMR-02	`BaseBo.Update()` Rev check opt-in; clinical records not protected	**P1**
N-04	Patient-context confirmation before high-impact actions	EMR-01	No confirmation gate when switching patients or performing irreversible actions	**P1**
N-05	Wrong-patient / wrong-context action rejection	EMR-01, ENC	Encounter scope not cross-validated against logged user's current patient context	**P1**
N-06	Allergy override audit trail	MED-01, F09	No override record when clinician overrides an allergy warning to prescribe	**P1**
N-07	Medication reconciliation at discharge	MED-04	`PatientDischargeMedication` model exists; reconciliation workflow not confirmed	**P2**
N-08	Nursing shift handover view	NUR-04	No shift handover screen; due doses not visible across shift boundary	**P2**

N-09	Abnormal result escalation with named recipient + read-back	ORD-03, LIS	No escalation chain in EMR result viewer	**P2**
N-10	Longitudinal patient timeline (encounters, notes, orders, results, meds)	EMR-01	No unified cross-encounter timeline view	**P2**
N-11	Clinical note provenance viewer (who wrote, signed, amended, when)	EMR-02	No provenance UI; audit fields exist but not surfaced	**P2**
N-12	Structured screening recalls / preventive care schedule	SPC (blueprint)	Immunization schedule exists; preventive care recalls absent	**P3**
N-13	Specialty activation governance (not all specialties enabled by default)	SPC	All specialty tabs always visible regardless of facility type	**P2**

5.5 Items Requiring Runtime or Clinical Validation

These items cannot be assessed from source code alone:

ID	Item	Why Runtime / Clinical Validation Needed
R-01	EMAR completeness — are due doses actually tracked?	`hims_emar` and `patientemar` tables need live data inspection; two tables with unclear purpose (OQ-04)
R-02	Prescription status transitions — what does `PrecriptionStatusId = 1/2/3` mean?	No enum definition found in source; requires ReferenceValue table query
R-03	`ProgressNoteStatusId = 2` = "Signed"?	Inferred from code; requires ReferenceValue master table query
R-04	`IsDirectBill = true` on PatientOrder — does it automatically bill?	Cannot trace from source alone; requires runtime test with billing enabled
R-05	Concurrent bed/stock/prescription scenarios	Requires synthetic test environment with multiple simultaneous sessions
R-06	Printing — does `PrintPrescription` produce correct output with doctor registration number?	Server-side PhantomJS rendering; requires live output inspection
R-07	PACS / radiology order flow — is there actual DICOM integration?	Source has PACS-related flags; no DICOM adapter code found; requires infrastructure check
R-08	Date correction (D-03) — does `ReviewDate` display correctly after fix?	Requires browser test in IST timezone
R-09	Blood bank compatibility / crossmatch workflow	`bloodrequest` model exists; compatibility check logic not found in source
R-10	Specialty clinical workflows (IVF, Obstetrics, Dialysis) — are they clinically validated?	Require specialty clinical owners to sign off field definitions and business rules
R-11	HL7/FHIR/ABDM integration — are any active?	Source mentions HL7 work; no ABDM/ABHA/NHCX implementation found; requires infra access

6. Revised Screen Inventory (v2)

All counts from Section 2 above. Refer to v1 specification for full state-by-state listing. This section adds status corrections from the blueprint comparison.

Functional Group Status Summary

Group	States	Backend APIs	React?	Blueprint Gap?
EMR Shell & Dashboard	6	PatientEMRDashboard	✗	N-10 (longitudinal timeline)
Consultation / Progress Notes	6	ConsultationRoute	✗	N-01, N-02, N-03 (signed lifecycle)
Doctor / Nursing / IVF Notes	10	DailyNote, ClinicalNotes	✗	N-01, N-02 (same lifecycle gap)
Vital Signs	4	PatientVitalRoute	✗	—
Diagnosis	3	PatientDiagnosisRoute	✗	P-09 (problem list)
Allergies	1	PatientAllergyRoute	✗	N-06 (override audit)
Prescriptions	11	PrescriptionRoute, Detail	❄️ Partial	D-01, D-02, D-03, P-01
Clinical Orders	7	PatientOrderRoute	❄️ List only	P-02, P-06, N-09
Procedure Orders	4	ProcedureOrderRoute	✗	P-06
EMAR	2	Emar + PatientEmar	✗	P-05, N-08
Nursing Charts	8	9 chart route files	✗	—
Patient History	4	Family/Condition/Immunization	✗	—
Consent & Documents	5	ClinicalDocument	✗	—
Discharge (in EMR)	7	Consultation discharge fields	✗	—
Specialty Workflows	14	14 specialty route files	✗	N-13 (governance)
Blood Bank	3	BloodRequestRoute	✗	R-09
Referral & Follow-up	5	—	✗	—
Financial Views (read)	3	—	✗	—
Task & Incident	3	IncidentReportingRoute	✗	—
Treatment Plans	2	TreatmentPlanRoute	✗	—
Total	**163**	**540**	**13 components**	**Multiple**

7. Developer Handoff Prompts (Updated)

H-01: Fix F09 — Prescription Action Buttons (DEFECT — do first)

CONTEXT : HIMS EMR – BUG FIX, not new migration

FILE : src/react-components/DoctorPrescribeFormScreen.tsx (lines 414–450)
public/views/emr/registration/prescriptions/doctorprescribe-form.js (lines 51, 1163)

DEFECT : canShowSaveBtn, canShowPrescribeBtn, canShowPrescribeOrderBtn are always falsy
→ Save and Prescribe buttons never appear in React UI
→ BUG 4: ismodal never assigned → Back/Save-as-Rx-Panel always shown

APPROACH:

1. In doctorprescribe-form.js (Angular controller): identify where these flags should be set based on user role + prescription status; ensure they are passed to reactProps correctly.
2. Server-side (PrescriptionRoute.ts): add status validation in Save/Update handlers: reject updates if PrescriptionStatusId is already 2 (finalized) or 3 (dispensed).
3. In DoctorPrescribeFormScreen.tsx: remove the BUG 4 comment; wire ismodal from currentcontext once the controller supplies it.
4. Test matrix: clinician with role=Doctor, statusId=0 → showSave=true; statusId=3 → showSave=false, showCancel=true, showPrint=true.

DO NOT : Change DB schema, add new endpoints, or change Angular routing.

ACCEPT : Authorized save/prescribe in each permitted state; unauthorized/finalized states block both UI and API calls.

H-02: Fix F08 — Date Loses One Day in IST

CONTEXT : HIMS EMR – BUG FIX, affects all React date fields under IST

FILES : src/react-components/DoctorPrescribeFormScreen.tsx:75–94
Any new React component using toISOString().slice(0,10)

DEFECT : date.toISOString().slice(0,10) converts to UTC before slicing → loses 1 day in UTC+5:30 (Asia/Kolkata)

FIX :

Replace this pattern:

```
const toDateInputValue = (d) => new Date(d).toISOString().slice(0, 10);
```

With:

```
const toDateInputValue = (d) => {
  const dt = d instanceof Date ? d : new Date(d);
  if (isNaN(dt.getTime())) return '';
  const y = dt.getFullYear();
  const m = String(dt.getMonth() + 1).padStart(2, '0');
  const day = String(dt.getDate()).padStart(2, '0');
  return `${y}-${m}-${day}`;
};
```

SCOPE : Fix in DoctorPrescribeFormScreen.tsx, then apply to ALL future EMR React components before they are merged.

TEST : With system timezone set to Asia/Kolkata (IST, UTC+5:30), selecting 2026-09-20 on the date picker should save 2026-09-20, not 2026-09-19.

H-03: Fix F10 — Signed Clinical Note Lifecycle (Backend Schema + Logic)

CONTEXT : HIMS EMR – NEW BACKEND FEATURE (F10 / EMR-02)

SCOPE : PatientClinicalNotes only (pilot; verify DailyNote separately)

CURRENT : PatientClinicalNotesBo.UpdatePatientClinicalNotes() overwrites in place.
No signed state; no amendment record.

REQUIRED CHANGES:

1. Schema additions to hims_patientclinicalnotes:

- SignedAt DATETIME nullable
- SignedBy BIGINT nullable FK-User
- SignedContent TEXT nullable – verbatim copy of note at signing
- NoteStatus INTEGER 1=Draft, 2=Signed, 3=Amended, 4=EnteredInError
- AmendmentOf BIGINT nullable FK-hims_patientclinicalnotes (self-ref)
- AmendmentReason STRING nullable

2. Update PatientClinicalNotesBo:

- AddPatientClinicalNotes: save with NoteStatus=1 (Draft)
- SignPatientClinicalNote: set NoteStatus=2, SignedAt=now(), SignedBy=userId, SignedContent=current note text. Reject if already Signed.
- UpdatePatientClinicalNotes: reject if NoteStatus=2 or 3. Return 403.
- AmendPatientClinicalNote: create new record with AmendmentOf=original.Id, NoteStatus=3, AmendmentReason required.
- DeletePatientClinicalNotes: reject if NoteStatus=2. Return 403.

3. Add routes: POST /EMR/SignPatientClinicalNote, /EMR/AmendPatientClinicalNote

DO NOT : Apply this schema to ALL note types simultaneously. Pilot on PatientClinicalNotes, then inventory DailyNote, PatientAnnotation, Consultation before extending.

ACCEPT : Signed content cannot be changed through any endpoint.
Amendments retain both versions and reasons.
Concurrent edits produce a conflict (after F11 fix).

RISK : Requires DB migration. Test rollback before deploying.

H-04: Fix F07 — React Bridge Nested Value Detection

CONTEXT : HIMS React bridge – BUG FIX, affects all React components

FILE : src/reactBridge.tsx:24-67

DEFECT : clonePropsSnapshot() uses shallow clone; isPropsSnapshotEqual() returns true when nested objects mutated in place.
Reproduction: obj.nested.value = 2 while prev.nested.value = 1
→ bridge reports no change → component does not re-render.

EMR IMPACT: Allergy alert (isPatientHasAllergy), prescription detail rows (nested drug objects), vital list updates all affected.

FIX : Replace shallow clone with deep clone in clonePropsSnapshot().
Option A: JSON.parse(JSON.stringify(obj)) – safe for plain objects
Option B: structuredClone(obj) – Node 17+; check runtime version
The reference report confirms BaseBo contains optional Rev checks – do not confuse concurrency (F11) with the bridge equality issue (F07).

TEST : Create a test that mutates a nested value in-place on the AngularJS scope and verify the React component re-renders.
Existing test stub (if any): api/src/Server/Modules/Registration/Router/PatientRoute.spec.ts (pattern example only).

H-05: Vital Signs React Migration (updated from v1)

CONTEXT : HIMS EMR – React migration (non-defect), AFTER H-01 through H-04 are done

SCREEN : patientemr.patientvitaltab (163 total EMR states; this covers 4)

APIs : POST /EMR/AddPatientVital, GetPatientVitals, ManagePatientVitals, PrintPatientVital, PrintPatientVitalWithoutHeader

DB TABLE: hims_patientvitals – PK: PatientVitalId

KEY FIELDS:

VitalId	FK-ClinicalMaster.VitalMaster
VitalValue / UOM	STRING – value and unit
LoincCode	STRING – informational only; do NOT validate or look up
ReferenceRangeFrom/To	STRING – display-only; NO clinical threshold enforcement

PerformedDate/By DATE / FK→User
 PatientVitalStatusId FK→ReferenceValue
 VitalQualifier STRING

SCOPE :

1. Tabular entry – multiple vitals at once (one row per vital type)
2. Current encounter vitals list
3. Trend chart – last 7–10 readings per vital type
4. Previous encounter vitals – read-only

DATE FIX: Use the pattern from H-02 for ALL date fields.

BRIDGE : Follow DoctorPrescribeFormScreen.tsx bridge. onAction only.

No direct fetch() from React. Verify nested prop fix (H-04)
 before binding vitals list to reactProps.

ACCEPT : Entry saves via AddPatientVital. Out-of-range highlighted via
 ReferenceRangeFrom/To display only. No clinical alert logic added.
 Date fields save the selected date correctly in IST.

H-06: Diagnosis Entry React Migration (updated)

CONTEXT : HIMS EMR – React migration, after H-01–H-04

SCREEN : patientemr.diagnosisstab (3 states)

API : POST /EMR/AddPatientDiagnosis, GetPatientDiagnosis, UpdatePatientDiagnosis

DB TABLE: patientdiagnosis – PK: PatientDiagnosisId

KEY FIELDS:

DiagnosisId FK→ClinicalMaster.Diagnosis
 Code STRING – ICD-10 code (display only; system does not validate)
 DiagnosisTypeId FK→ReferenceValue (Primary/Secondary/Complication)
 DiagnosisStatusId FK→ReferenceValue (Active/Resolved/Chronic)
 ConditionTypeId / ConditionStatusId FK→ReferenceValue
 DOA/DOD DATE – use H-02 date fix

SCOPE :

1. Diagnosis typeahead search + manual entry
2. Add/edit form with type, status, condition fields
3. Current encounter diagnosis list
4. History – all diagnoses across encounters (filter PatientId only)

DO NOT : Merge PatientDiagnosis and PatientCondition into one UI element
 without clinical owner approval. Blueprint (P-09) flags this as
 a future structured problem list – not in this batch.

ACCEPT : Diagnosis with ICD-10 code saves and displays correctly.
 Edit of a signed note via diagnosis tab does not bypass F10 controls.

H-07: Allergy Entry + Persistent Alert Banner

CONTEXT : HIMS EMR – React migration + shared component (P1, N-06)

SCREEN : patientemr.patientallergy (1 state)

API : POST /EMR/AddPatientAllergy, GetPatientAllergys,
 UpdatePatientAllergy, DeletePatientAllergy

DB TABLE: patientallergies – PK: PatientAllergyId

KEY FIELDS:

AllergyTypeId FK→ReferenceValue (Drug/Food/Environmental/Other)
 AllergySeverityId FK→ReferenceValue (Mild/Moderate/Severe/Life-threatening)
 ADRStatus / ADRScoreId
 StartDate/EndDate DATE – use H-02 date fix
 PerformedDate/By DATE/FK→User

SCOPE :

1. Allergy list – severity colour-coded (Severe/Life-threatening = red)

2. Add/edit form
3. Delete requires confirmation + reason; soft-delete only (Status=0)
4. Shared banner component: persistent across all EMR tabs when
isPatientHasAllergy = true. Bridge fix (H-04) required first.
5. Banner clears when patient context switches (verify with bridge fix)

OVERRIDE AUDIT (N-06):

When a drug matching a known allergy is prescribed and the clinician chooses to proceed: log an override record to a new table or to PatientAllergy with OverrideBy / OverrideAt / OverrideReason fields.

Do NOT block the prescription save – only audit the override.

DO NOT : Compute drug-interaction calculations. Only match allergy name to drug name (string comparison). Clinical lead must define matching rules; do not invent them.

ACCEPT : Allergy banner visible on all EMR tabs when allergies exist.

Banner updates correctly when new allergy is added (requires H-04).

Delete is auditable (UpdatedBy/At preserved).

8. Implementation Backlog (v2 — Prioritised)

Phase 0 — Security and Build Blockers (Gate A prerequisite)

Backlog ID	Deliverable	Owner	Dependencies	Acceptance
B0-01	Fix PrivilegeHelper — all 435 methods returning false; fix 7 duplicate keys	Frontend	None	EMR screens gate actions by role
B0-02	Fix CORS wildcard (restrict to known origins)	Backend/security	None	Non-whitelisted origin → 403
B0-03	Move Bearer token from localStorage to httpOnly cookie	Backend/security	None	Token not readable from JS
B0-04	Create `api/.env` from template	DevOps	None	Application starts on LAN
B0-05	Fix TypeScript compile (7 duplicate keys, 14 TS2393, 8 TS6133)	Frontend	B0-01	`tsc --noEmit` exits 0
B0-06	Fix F07: React bridge deep clone	Frontend	B0-05	Nested prop change → re-render
B0-07	Fix F08: Date timezone (IST -1 day)	Frontend	None	Date fields correct in IST
B0-08	Fix F09: Prescription action matrix	Frontend+Backend	B0-06, B0-07	Save/Prescribe visible and server-validated
B0-09	Patient resource authorization on Get* endpoints	Backend	None	Cross-facility read → 403

Phase 1 — Clinical Safety Fixes (Gate B prerequisite)

Backlog ID	Deliverable	Owner	Dependencies	Acceptance
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B1-01	Signed note lifecycle for PatientClinicalNotes (F10)	Backend	DB migration	Signed note immutable; amendment creates linked record
B1-02	Enforce Rev concurrency on clinical records (F11)	Backend	B1-01	Concurrent edit with stale Rev → 409
B1-03	Patient context confirmation gate for high-impact actions	Frontend	B0-06	Wrong-patient prescription blocked by UI + server
B1-04	Allergy override audit record	Backend	B1-01	Override logged with OverrideBy/At/Reason
B1-05	Document and publish ProgressNoteStatusId enum values	Backend/DBA	ReferenceValue table query	Enum table published in docs
B1-06	Clarify and consolidate dual EMAR tables (OQ-04)	Backend/DBA	None	Single canonical EMAR source or documented split
B1-07	Synthetic test harness (F13)	QA	B0-05	CI runs type-check + meaningful API tests on each push

Phase 2 — Core Clinical React Migration (Gate B)

Backlog ID	Screen	Handoff	Dependencies	Acceptance
B2-01	Vital Signs form + chart	H-05	B0-06, B0-07	Entry saves; chart renders; IST dates correct
B2-02	Allergy form + persistent banner	H-07	B0-06, B1-04	Banner visible; override audited
B2-03	Diagnosis entry + history	H-06	B0-06, B0-07	ICD-10 code preserved
B2-04	Prescription drug line items	New handoff needed	B0-08	Drug entry saves; allergy context shown
B2-05	Clinical order entry form	New handoff needed	B0-06	Order saves; pending list updates
B2-06	Patient identity banner	New handoff needed	B0-06	MRN/name/allergy visible on all EMR tabs

Phase 3 — Nursing and EMAR (Gate C)

Backlog ID	Screen	Dependencies
B3-01	EMAR administration screen (dose/time/witness)	B1-06, B1-02
B3-02	Nursing shift handover view	B3-01
B3-03	Intake/Output chart React	B0-06
B3-04	Nursing notes editor	B1-01 pattern applied

Phase 4 — Specialty and Longitudinal (Gate D)

Backlog ID	Screen	Dependencies
B4-01	Longitudinal patient timeline	B2-01–B2-06
B4-02	Specialty governance (configurable activation)	FacilityPreference
B4-03	Dental, Ophthalmology, Physiotherapy React	B0-06
B4-04	Obstetrics / IVF React	B4-03
B4-05	Country pack architecture	B0-02, B0-03

9. Test Plan (v2)

9.1 Pilot Journey Tests (Gate B exit criteria)

Per the reference document, Gate B requires these journeys to pass with synthetic data:

1. Registration → Encounter → Note → Order → Prescription → Bill → Follow-up

- Verify: no duplicate patient, stable encounter identity, prescription saved with correct date, order reaches pending list, bill created

2. Allergy → Prescription → Pharmacy → Dispense → Administration → Reconciliation

- Verify: allergy warning appears, override audited, correct drug/dose reaches EMAR once, concurrent stock does not double-dispense

3. Signed Note → Amendment → View Provenance

- Verify: note content immutable after sign; amendment creates linked record; both visible

4. Concurrent Edit Race

- Verify: two simultaneous updates with same Rev → first succeeds, second returns 409

9.2 Explicit Completion Criteria (per backlog item)

A backlog item is **complete** when ALL of:

- [] API endpoints return correct status for valid and invalid inputs
- [] No previously passing tests regress
- [] React component renders without console errors
- [] Soft-delete: deleted records absent from all `Get*` responses
- [] Audit fields (`CreatedBy/At` , `UpdatedBy/At`) set on every create/update
- [] Privilege check gates screen actions (after B0-01)

- [] Date fields save and display correctly in IST (after B0-07)
- [] No direct `fetch()` from React — all via AngularJS controller callbacks
- [] Allergy banner visible when allergies exist (after B0-06)

10. Delivery Gates (from Reference Document)

Gate	Name	EMR Exit Criteria
A	Authentication and authorization baseline	F01–F05, B0-01–B0-09 complete; EMR endpoints return 403 for unauthorized requests
B	Pilot clinical workflows	F07–F11 resolved; prescription and note lifecycle working; pilot journey tests pass
C	Hospital and financial integrity	EMAR completeness, concurrent edit, nursing handover; signed-note amendment in production
D	Controlled expansion	Specialty governance, country pack, longitudinal timeline; one additional country validated

11. Open Questions (Updated)

ID	Question	Impact	Resolution Status
OQ-01	Blueprint PDF full text (EMR-01–EMR-05 page-level contracts)	 Critical	✅ **RESOLVED** — `pdftotext` (poppler 26.09.0) installed; full text extracted. See Section 13.
OQ-02	Radiology: is DICOM/PACS integration active or planned?	 High	RIS module mapped as P2 in blueprint; PACS flag exists; no DICOM adapter code found
OQ-03	Audit log table for allergy/diagnosis changes	 High	No dedicated table found; B1-04 adds override audit
OQ-04	Why two EMAR tables? (`hims_emar` vs `patientemar`)	 Medium	B1-06 resolves
OQ-05	`PhysiotherapyTreatement` typo — intentional?	 Low	Unresolved
OQ-06	`BaseBo.Update()` Rev check — opt-in or enforced?	✅ **Resolved**	Reference doc confirms: "optional Rev concurrency checks" — opt-in
OQ-07	Signed note amendment workflow	✅ **Resolved**	F10 analysis + H-03 handoff + EMR-02 blueprint contract define the required implementation
OQ-08	`PatientOrder.IsDirectBill` auto-creates billing?	 High	R-04 — requires runtime test

OQ-09	`LoincCode` on PatientVital — from master or manual?	 Medium	Runtime data inspection required
OQ-10	International expansion scope and chosen next country	 High	UAE/Nepal variants confirmed in source; next country selection deferred to product decision
OQ-11	`ProgressNoteStatusId` enum values	 **Resolved by inference**	`= 2` triggers finalization (ConsultationBo confirmed); B1-05 formalises
OQ-12	SIMPLEX staging review — what do the 30 EMR screen captures show?	 High	RG-02 — captures marked "not preserved"; need recovery
OQ-13	ABDM/ABHA/NHCX/FHIR — are any external integrations active?	 High	Reference doc: "no ABDM/ABHA/NHCX/FHIR implementation found in bounded search"

12. Residual Gaps — Why This Specification Is Not Labelled Complete

Gap ID	Description	Status	How to Resolve
~**RG-01**~	~Blueprint PDF page-level contracts not extracted~	 **CLOSED**	Resolved via pdftotext + poppler 26.09.0
RG-02	SIMPLEX staging screen captures for ~30 EMR screens are "not preserved"	 Open	Recover the browser captures from the SIMPLEX staging session
RG-03	Runtime validation of 11 items (R-01–R-11) not possible without a synthetic test environment	 Open	Establish synthetic DB per F13 (B1-07)
RG-04	`PrecriptionStatusId` and `ProgressNoteStatusId` reference values not queried from DB	 Open	Run: `SELECT * FROM referencevalue WHERE Category='PrescriptionStatus'`
RG-05	EMAR completeness (R-01) — two tables, purpose unclear	 Open	B1-06 + runtime inspection of live EMAR data

13. Blueprint Contract Reference (Extracted from PDF)

Source: `healthcare-platform-full-blueprint.pdf` v1.0, 20 September 2026

Extracted with pdftotext (poppler 26.09.0) on 2026-09-21.

EMR — Clinical Chart and Documentation

Contract	Name	Key Inputs	Actions
----------	------	------------	---------

EMR-01	Longitudinal chart	Encounter, timeline filters, problems, allergies, medicines	Review prior care, mark reviewed information, inspect source
EMR-02	Consultation note	Complaint, history, examination, assessment, plan	Save draft, compare edits, sign, append correction
EMR-03	Problem and allergy lists	Code, display term, onset, severity, verification, status	Add or reconcile entries, mark resolved, record uncertainty
EMR-04	Template and specialty forms	Template version, field rules, terminology bindings	Preview, clinically approve, publish effective version
EMR-05	Clinical inbox	Results, referrals, unsigned notes, follow-up tasks	Assign responsibility, acknowledge, act, escalate overdue work

Blueprint Workflow (EMR):

Draft → reviewed → signed → amended. Signed content is immutable; amendments are separately signed and linked. Imported records retain origin and verification status.

EMR-F-01 Mandatory Rules (from blueprint):

Show patient identity and encounter continuously. Do not equate no allergy information with no known allergies. Never silently overwrite concurrent edits or copy stale findings as newly observed.

EMR-A-01 Acceptance:

Two editors receive a conflict resolution view; a signed note cannot be overwritten. An imported allergy retains its source. Measure unsigned notes and inbox items beyond agreed response time.

EMR-02 Detailed Contract (from blueprint page 49):

Aspect	Blueprint Requirement	HIMS Current State	Gap?
Actor	Authorised clinician in a valid encounter	No server-side encounter scope check	✗ Gap — OQ-05/F05
Identity header	Patient identity and encounter continuously visible	Shell exists; no persistent banner in React	✗ Gap — N-04, P-11
Template version	Template version and author role explicit	Not found in PatientClinicalNotes model	✗ Gap
Imported content	Labelled with source and date	No import provenance fields	✗ Gap
Draft save	Save draft revisions with concurrency version	Rev field exists; not enforced	⚠ Partial — F11
Signing rules	Recheck authority, credentials, encounter identity; freeze signed content	No signing lifecycle	✗ Gap — F10, N-01
Signing output	Preserve exact signed content, author, time	No SignedContent/SignedAt/SignedBy columns	✗ Gap — F10, B1-01

Amendment	Records reason and linked signed version	No AmendmentOf FK; no amendment record	✗ Gap — N-02, H-03
Idempotency	Reconcile operation ID before retry on lost response	No idempotency key on note endpoints	✗ Gap
Concurrent edit	Display differences; do not prefer last writer	Last-write-wins currently	✗ Gap — F11, B1-02
Proposed API	POST /v1/notes/{id}/sign, POST /v1/notes/{id}/amendments	Not yet implemented	✗ Gap — B1-01

MED — Prescribing and Medication Safety

Contract	Name	Key Inputs	Actions
MED-01	Medication reconciliation	Reported medicines, prior orders, source, current status	Confirm, discontinue with reason, flag uncertain entries
MED-02	Prescription composer	Drug, formulation, strength, dose, units, route, frequency, duration	Validate completeness, review safety advice, sign prescription
MED-03	Safety review	Allergy, interaction, duplicate therapy, relevant clinical context	Show evidence and severity, resolve or justify permitted overrides
MED-04	Prescription issue and amendments	Signed version, pharmacy destination, delivery status	Issue document, send through approved channel, revoke or amend
MED-05	Formulary administration	Drug identifiers, substitutions, restrictions, effective dates	Review catalogue updates, publish version, manage shortage alternatives

Blueprint Workflow (MED):

Draft → signed → active → completed, stopped or cancelled; dispensing and administration use separate records.

MED-F-01 Mandatory Rules (from blueprint):

Clinical rules and reference data require qualified governance. Missing weight or renal data is not silently interpreted as normal. Units must be explicit; substitution needs local policy and authorised review.

MED-02 Detailed Contract (from blueprint page 50):

Aspect	Blueprint Requirement	HIMS Current State	Gap?
Context	Current allergies, reconciliation state, measurements visible	`isPatientHasAllergy` flag; no reconciliation state	⚠️ Partial
Order fields	Drug, formulation, strength, dose, units, route, frequency, duration, quantity, indication, as-needed reason, instructions, prescriber, order version	Most fields exist in `hims_prescriptiondetails`	✅ Mostly covered

Clinical validation	Validate dimensional consistency deterministically	No server-side dose validation	✗ Gap — clinical lead required
Knowledge rules	Invoke approved knowledge rules only with applicable population	No drug knowledge adapter	✗ Gap — MED-F-01
Override	Clinical reviewers define severity, blocking behaviour, permitted overrides	No override governance	✗ Gap — N-06
Signing output	Creates medication order + downstream pharmacy task	Prescription saves; pharmacy integration unverified	⚠ R-04
Amendment	Changed order linked to predecessor; pharmacists/nurses notified	No predecessor FK; no notification	✗ Gap — MED-04
Cancellation	Does not erase historical administrations	Soft-delete via Status=0	✓ Preserved

ORD — Orders and Result Follow-up

Contract	Name	Key Inputs	Actions
ORD-01	Order composer	Patient, encounter, service, indication, priority, timing	Validate prerequisites, sign order, route to destination
ORD-02	Order tracking	Service state, performer, expected completion, exceptions	Review progress, cancel with reason, escalate delays
ORD-03	Result viewer	Result version, units, reference interval, specimen and source	Compare trends, acknowledge, add action and patient follow-up
ORD-04	Critical result worklist	Critical flag, recipient, contact attempts, acknowledgement	Notify, document read-back under policy, escalate to backup
ORD-05	Order sets	—	—

HIMS Current State vs ORD:

- ORD-01: `PatientOrderRoute` + order entry form (Angular) — partial ✓
- ORD-02: `ClinicalPendingOrdersListScreen` (React) + `OrderTrackerScreen` (React) — partial ⚙
- ORD-03: Lab result viewer exists in Angular — ✓ (unverified for trend comparison)
- ORD-04: `PatientCriticalOrder` route in LIS — ⚠ critical result worklist not in EMR
- ORD-05: `favorders` state exists — ? unverified

NUR — Nursing and Medication Administration

Contract	Name	Key Inputs	Actions
NUR-01	Ward patient list	—	—

NUR-02	Flowsheets and care plans	—	—
NUR-03	Medication administration record	—	—
NUR-04	Nursing tasks and handover	—	—
NUR-05	Intake output and devices	—	—

HIMS Current State vs NUR:

- NUR-01: `NursingDashboardComponent` (React) 
- NUR-02: Nursing charts in Angular (BP, I/O, Ventilator)  (unverified depth)
- NUR-03: EMAR routes exist — two tables (OQ-04) 
- NUR-04: No shift handover screen confirmed  Gap — N-08
- NUR-05: `hims_intakeoutputchart` model + route 

Read-only analysis. No application code was modified.

All claims marked [PROPOSAL] in v1 have been replaced with verified findings where confirmed.

Do not start implementation of B1-01 (signed note lifecycle — DB migration) without clinical lead approval and a tested rollback plan.

This specification is IN PROGRESS until RG-02 (SIMPLEX screen captures) is resolved.

OQ-02	Radiology: is DICOM/PACS integration active or planned?	 High	RIS module mapped as P2 in reference; no DICOM code found
OQ-03	Audit log table for allergy/diagnosis changes	 High	No dedicated table found; B1-04 adds override audit
OQ-04	Why two EMAR tables? (`hims_emar` vs `patientemar`)	 Medium	B1-06 resolves
OQ-05	`PhysiotherapyTreatement` typo — intentional?	 Low	Unresolved
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RG-03	Runtime validation of 10 items (R-01–R-11) not possible without a synthetic test environment	Establish synthetic DB per F13 (B1-07)
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RG-05	EMAR completeness (R-01) — two tables, purpose unclear	B1-06 + runtime inspection of live EMAR data

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- *All claims marked [PROPOSAL] in v1 have been replaced with verified findings where possible.*
- *Do not start implementation of B1-01 (signed note lifecycle — DB migration) without clinical lead approval and a tested rollback plan.*